

# EFX<sub>0</sub> Allocations with at Most Three Relevant Goods: A Short Proof by Exchange Cycles

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**Abstract.** An allocation of indivisible goods is EFX<sub>0</sub> if no agent envies another agent’s bundle after removing any single good from it, even one the envious agent values at zero. We give a short proof that every instance with nonnegative additive valuations in which each agent positively values at most three goods has a complete EFX<sub>0</sub> allocation, in which moreover all bundles but at most one have at most two goods, and a polynomial-time algorithm, Draft and Exchange (DE), that computes one with at most  $4n$  exchanges. The heart of the proof is an *Improvement Lemma*: a valid state from which no free agent can absorb the leftover goods is Pareto-dominated by another valid state, reached by one exchange along a cycle or a path of agents. Existence, and the shape for agents that value exactly three goods and are balanced, were machine-checked in Lean in previous work, by a much longer construction. The new proof is a written proof, derived independently twice, refereed once with no error found (by an independent AI agent; there has been no human peer review), and checked by computer on millions of states; it is not machine-checked.

**Keywords:** Fair division · Indivisible goods · EFX · Exchange cycles

## 1 Introduction

Agents  $N$  ( $|N| = n \geq 1$ ) share a set  $M$  of  $m$  indivisible goods with nonnegative additive valuations  $v_i(S) = \sum_{g \in S} v_i(g)$ . An allocation  $X$  assigns every good to exactly one agent, who receives the bundle  $X_i$ ; no good is left out (allocations are *complete*). It is EFX<sub>0</sub> if

$$v_i(X_i) \geq v_i(X_j \setminus \{h\}) \quad \text{for all } i \neq j \text{ and all } h \in X_j,$$

and EFX asks this only for goods  $h$  with  $v_i(h) > 0$  [4]. So EFX<sub>0</sub> implies EFX, and the two coincide for positive values. *Under EFX<sub>0</sub>, if  $X_j$  contains a good that  $i$  values at 0, then  $i$  must not envy  $X_j$  at all.* Whether complete EFX allocations always exist for additive valuations is open; they are known to exist in special cases, among them three agents [5] or three types [7],  $m \leq n + 3$  [15], four agents and at most nine goods [3], goods

valued by at most two agents [1,6,16,8], and hypergraphs of girth at least four [9] (Table 1 in Appendix F). Let  $R_i = \{g : v_i(g) > 0\}$ . If  $|R_i| \leq 2$  for all  $i$ , serial dictatorship gives  $\text{EFX}_0$  [13]. For  $|R_i| \leq 4$ , Viswanathan and Mehta [17] announce ordinary EFX, with a proof sketch; their algorithm gives the goods nobody values to an arbitrary agent, which can break  $\text{EFX}_0$ .

**Theorem 1 (existence and shape).** *Every instance with  $n \geq 1$  agents and nonnegative additive valuations in which every agent positively values at most three goods has a complete  $\text{EFX}_0$  allocation in which all bundles but at most one have at most two goods. In particular this holds when every agent values exactly three goods and is balanced (values none of them more than the other two together).*

**Theorem 2 (algorithm).** *Algorithm DE (Section 5) computes such an allocation with at most  $n$  peeling steps and at most  $4n$  exchanges, each taking time polynomial in  $n$  and  $m$ ; on real values, a comparison of two sums of at most two values of one agent counts as one step, as in [10].*

*Previous work and status.* Theorem 1 (existence, and the shape for agents that value exactly three goods and are balanced) was proved in previous work [10] with a construction, K3ALG, whose proof uses blocks of a draft, leaders, an owner test and one rotation; existence and that shape are machine-checked in Lean 4 for values in any linearly ordered cancellative commutative monoid, and K3ALG with its operation bound over  $\mathbb{N}$  (over ordered values in a comparison model) [10,12]. This paper gives a much shorter proof. Its core, the Improvement Lemma (Theorem 4), was derived independently twice, by two different arguments [11]. It was refereed once, with no error found, by an independent AI agent (a separate session of the coding assistant that helped with this work, given only the written proof); there has been no human peer review. Every step has been checked by computer as described in Section 6; it is not machine-checked. A long version with every proof in the body, a worked example with a figure, and pseudocode is `paper/k3-simple/long.pdf` in [11]. Soundness (Theorem 3), the correctness of DE and the shape for all instances with at most three relevant goods per agent are likewise written proofs, checked by computer and not machine-checked.

## 2 Preliminaries

The *threat* of a nonempty set  $B$  to agent  $i$  is  $\theta_i(B) = v_i(B) - \min_{h \in B} v_i(h)$ , and  $\theta_i(\emptyset) = 0$ ; agent  $i$  is *safe* in  $X$  if  $v_i(X_i) \geq \theta_i(X_j)$  for all  $j \neq i$ , so  $X$

is EFX<sub>0</sub> iff every agent is safe. A good is *alone* if it is the only good of its bundle. When  $|R_i| = 3$ , we name  $i$ 's goods so that  $v_i(a_i) \geq v_i(b_i) \geq v_i(c_i) > 0$ , ties broken by index, and write  $a_i \succ_i b_i \succ_i c_i$ . Agent  $i$  is *balanced* if  $v_i(a_i) \leq v_i(b_i) + v_i(c_i)$ , *strictly* if the inequality is strict. Proofs of this section are in Appendix A.

**Lemma 1 (threats).** (a)  $\theta_i(B) = 0$  if  $|B| \leq 1$ ; (b)  $\theta_i(B) \leq v_i(B \cap R_i)$ ; (c)  $\theta_i(\{g, h\}) = \max(v_i(g), v_i(h))$ .

**Lemma 2 (safety).** Let  $i$  be balanced with  $|R_i| = 3$ , and  $X$  an allocation. (a) If  $b_i, c_i \in X_i$ ,  $i$  is safe. (b) Let  $y \in X_i \cap R_i$  or  $y = \perp$ , and  $G_y = \{g \in R_i : g \succ_i y\}$  ( $G_\perp = R_i$ ). If no bundle  $X_j$ ,  $j \neq i$ ,  $|X_j| \geq 2$ , contains a good of  $G_y$ , then  $i$  is safe, unless  $y = a_i$  and some  $X_j$ ,  $j \neq i$ ,  $|X_j| \geq 3$ , contains  $b_i$  and  $c_i$ .

**Lemma 3 (peeling).** Let  $i \in N$  and  $P \subseteq M$  with  $|P| \leq 1$  and  $v_i(P) \geq v_i(M \setminus P)$ . If the instance without  $i$  and  $P$  has an EFX<sub>0</sub> allocation  $Z$ , then  $Z$  with  $X_i = P$  is EFX<sub>0</sub>.

*Rule R1:* with  $G$  the remaining goods and  $p$  agent  $i$ 's favourite good of  $G$ , agent  $i$  leaves with  $P = \emptyset$  if it values no good of  $G$ , and with  $P = \{p\}$  if  $v_i(G \setminus \{p\}) \leq v_i(p)$ .

**Lemma 4 (when R1 applies to nobody).** If  $i$  values at most three goods of  $G$  and R1 does not apply to  $i$ , then  $i$  values exactly three goods of  $G$  and is strictly balanced with respect to them.

### 3 States and Soundness

In the *core case* every agent values exactly three goods and is strictly balanced; goods valued by nobody may exist. A *state*  $Y$  gives each agent  $i$  a holding  $Y_i \in \{\emptyset, \{a_i\}, \{b_i\}, \{c_i\}, \{b_i, c_i\}\}$ , pairwise disjoint.  $U$  is the set of *pair holders* ( $Y_i = \{b_i, c_i\}$ );  $J = M \setminus \bigcup_i Y_i$  is the set of *leftover* goods; for  $i \notin U$  with  $|Y_i| = 1$  we write  $Y_i = \{y_i\}$ . The *utility*  $u_i(Y)$  is 4, 3, 2, 1, 0 for the pair,  $\{a_i\}$ ,  $\{b_i\}$ ,  $\{c_i\}$ ,  $\emptyset$ ;  $Y'$  *Pareto-dominates*  $Y$  if  $u_i(Y') \geq u_i(Y)$  for all  $i$ , strictly for some  $i$ .

- The *needs* are  $N_i = \emptyset$  for  $i \in U$ ,  $N_i = R_i$  if  $Y_i = \emptyset$ , and  $N_i = \{g \in R_i : g \succ_i y_i\}$  otherwise;  $NA = \bigcup_i N_i$ . No agent needs its own good. ( $N$  is the set of agents;  $N_i$ , with a subscript, are the needs of agent  $i$ .)
- $Y$  is *valid* if every  $g \in NA$  is the only good of some agent ( $Y_j = \{g\}$ ; then  $j \notin U$ ).

- An agent  $i \notin U$  is *free* if  $Y_i = \emptyset$  or  $y_i \notin NA$ ;  $F$  is the set of free agents.
- The *need digraph*  $D$  has the agents outside  $U$  as vertices and an arc  $j \rightarrow j'$  when  $j$  holds a single good  $y_j \in N_{j'}$ . It has no loops.
- Agent  $x$  is *exposed* for  $o$  if  $x \neq o$ ,  $x \notin U$ ,  $Y_x = \{a_x\}$  and  $b_x, c_x \in J \cup Y_o$ ;  $E_o$  is the set of these agents.
- Agent  $o$  is a *valid absorber* with set  $H$  if  $o \in F \cup U$ , (A1)  $\{b_x, c_x\} \cap J \neq \emptyset$  for every  $x \in E_o$ , and (A2)  $H \subseteq J$  meets every  $\{b_x, c_x\} \cap J$  ( $x \in E_o$ ) and  $|H| \leq |F \setminus \{o\}|$ .  $Y$  is *completable* if some agent is a valid absorber.
- The *completion* with  $o$  and  $H$ : each good of  $H$  goes to a different free agent other than  $o$ ;  $o$  receives  $X_o = Y_o \cup (J \setminus H)$ ; every other agent  $i$  keeps  $X_i = Y_i$ , or  $X_i = Y_i \cup \{h\}$  if it received  $h \in H$ .

**Theorem 3 (soundness).** *If  $Y$  is valid and  $o$  is a valid absorber with set  $H$ , the completion is  $EFX_0$ , and only  $X_o$  can have more than two goods.*

The proof (Appendix B) shows that needed goods stay alone, and that no agent  $x \neq o$ ,  $x \notin U$ , holding only  $a_x$  finds  $b_x$  and  $c_x$  in  $X_o$ : otherwise  $b_x, c_x \in J \cup Y_o$ , so  $x \in E_o$ , and  $H$  took one of them out of  $X_o$ . Lemma 2 then applies to every agent.

**Lemma 5.** (Draft) *If the agents, in any order, each take the first of  $a_i, b_i, c_i$  not yet taken (or nothing), the result is a valid state. Now let  $Y$  be valid and  $Y'$  a state. (F1) If  $u_i(Y') \geq u_i(Y)$ , then  $N'_i \subseteq N_i$ . (F2) Every agent outside  $U \cup F$  has an out-arc in  $D$ . (Transfer) If a state  $Y'$  has  $u_i(Y') \geq u_i(Y)$  for all  $i$ , strictly for some, and every  $g \in NA$  is the only good of some agent in  $Y'$ , then  $Y'$  is valid and Pareto-dominates  $Y$ .*

*Proof.* (Draft) A good that  $i$  needs was taken before  $i$ 's turn, alone, by an earlier agent, and holdings do not change. (F1) If  $Y'_i$  is the pair,  $N'_i = \emptyset$ ; if  $Y'_i = \{g'\}$ , then  $Y_i$  is  $\emptyset$  or a single good ranked at most as high as  $g'$ , so every good ranked above  $g'$  is in  $N_i$ ; if  $Y'_i = \emptyset$ , then  $Y_i = \emptyset$ . (F2) Such a  $j$  holds  $y_j \in NA$ , so some  $j'$  needs  $y_j$ ;  $j' \neq j$ , and  $j' \notin U$  since pair holders need nothing. (Transfer) By (F1),  $NA' \subseteq NA$ .  $\square$

## 4 The Improvement Lemma

**Theorem 4 (Improvement Lemma).** *In the core case, let  $Y$  be valid, with no free agent a valid absorber and some agent not in  $U$ . Then some valid state Pareto-dominates  $Y$ , obtained by one need cycle, pair chain or exchange cycle, found in polynomial time.*

If  $Y$  is not completable, the hypotheses hold (if  $U = N$ , every agent is a valid absorber with  $H = \emptyset$ , as exposed agents are outside  $U$ ). So every valid state that no valid state dominates is completable. In the proof,  $Y$  is valid.

**Lemma 6 (need cycle; pair chain).** (a) If  $D$  has a cycle,  $Y$  is dominated by a valid state. (b) If some  $x \notin U$  has  $Y_x = \{a_x\}$  and  $b_x, c_x \in J$ ,  $Y$  is dominated by a valid state.

*Proof.* (a) Let  $j_0 \rightarrow \dots \rightarrow j_{k-1} \rightarrow j_0$  be a cycle ( $k \geq 2$ ); its agents hold single goods. Let each  $j_{t+1}$  take  $y_{j_t}$  (indices mod  $k$ ). The goods are permuted, each  $j_{t+1}$  receives a good it needs, and a needed good held by  $j_t$  is now held alone by  $j_{t+1}$ ; others are unchanged. Transfer applies. (b) Walk  $j_0 = x$ ,  $j_{t+1}$  = an out-neighbour of  $j_t$ , while one exists. A repeat gives a cycle: use (a). Otherwise the walk stops at  $j_k$  with no out-arc. Let  $x$  take  $\{b_x, c_x\}$  and  $j_t$  take  $y_{j_{t-1}}$  for  $1 \leq t \leq k$  ( $y_{j_0} = a_x$ ); the old good of  $j_k$ , if any, becomes leftover. Holdings stay disjoint ( $b_x, c_x$  were leftover),  $x$  goes from 3 to 4, and each  $j_t$ ,  $t \geq 1$ , gets a needed good. Leftover goods are not needed (validity);  $y_{j_t}$  ( $t < k$ ) moves to  $j_{t+1}$  alone; and  $j_k \notin U$  has no out-arc, so  $j_k \in F$  by (F2) and its old good is not needed. Transfer applies.  $\square$

From now on assume **(P)**: no  $x \notin U$  with  $Y_x = \{a_x\}$  has  $b_x, c_x \in J$ .

**Lemma 7.** Assume (P). (a) If  $o \in F$  holds nothing,  $E_o = \emptyset$  and  $o$  is a valid absorber with  $H = \emptyset$ . (b) If  $o \in F$  holds  $y_o$ , every  $x \in E_o$  has  $\{b_x, c_x\} = \{y_o, h_x\}$  with  $h_x \in J$ ; with  $H_o = \{h_x : x \in E_o\}$ ,  $o$  is a valid absorber iff  $|H_o| \leq |F| - 1$ , and then with  $H = H_o$ . (c) An agent  $x$  lies in  $E_o$  for at most one free  $o$  holding a good, so  $h_x$  depends only on  $x$ . (d) If  $F = \emptyset$  and  $U \neq N$ ,  $D$  has a cycle.

*Proof.* (a)  $x \in E_o$  would have  $b_x, c_x \in J$ , against (P). (b)  $b_x, c_x \in J \cup \{y_o\}$ , not both in  $J$ , and  $b_x \neq c_x$ ; so one is  $y_o$  and the other is leftover. Then  $\{b_x, c_x\} \cap J = \{h_x\}$ , and  $H$  meets all of these iff  $H \supseteq H_o$ ; and  $|F \setminus \{o\}| = |F| - 1$ . (c) If  $x \in E_o \cap E_{o'}$ ,  $o \neq o'$ , then  $\{b_x, c_x\} = \{y_o, y_{o'}\}$  has no leftover good, against (b). (d) Every agent outside  $U$  has an out-arc by (F2), and there is one.  $\square$

The *exchange digraph*  $D^+$  (under (P)) adds to  $D$  an *exposure arc*  $o \rightarrow x$ , labelled  $h_x$ , for every  $o \in F$  holding a good and every  $x \in E_o$ . Heads of exposure arcs hold their tops and need nothing; heads of need arcs need a good; so the head determines the kind of arc. The *exchange* along a

cycle  $(w_0, \dots, w_{k-1})$  of  $D^+$  ( $k \geq 2$ , indices mod  $k$ ) lets  $w_{t+1}$  take its pair  $\{y_{w_t}, h_{w_{t+1}}\}$  if  $w_t \rightarrow w_{t+1}$  is an exposure arc, and  $y_{w_t}$  alone if it is a need arc; agents off the cycle keep their holdings. Every cycle agent holds a single good and has one predecessor, so this is well defined.

**Lemma 8 (exchange cycle).** *Assume (P). If the exposure arcs of a cycle  $C$  of  $D^+$  have distinct labels, the exchange along  $C$  is valid and Pareto-dominates  $Y$ ; its new pair holders are the heads of the exposure arcs.*

*Proof.* Each  $y_{w_t}$  moves to  $w_{t+1}$  only; the labels are leftover and distinct; agents off  $C$  keep other goods. So holdings are disjoint. Exposure heads go from 3 to 4; need heads receive a needed good; others are unchanged. If  $g \in NA$  is held by  $j$  alone and  $j = w_t \in C$ , then  $w_t \notin F$ , so  $w_t \rightarrow w_{t+1}$  is a need arc and  $w_{t+1}$  holds  $g$  alone; if  $j \notin C$ ,  $j$  keeps  $g$ . Transfer applies.  $\square$

**Lemma 9 (an exchange cycle exists).** *Assume (P),  $F \neq \emptyset$ , no free agent holds nothing, and  $|H_o| \geq |F|$  for all  $o \in F$ . Then  $D^+$  has a cycle whose exposure arcs have distinct labels.*

*Proof.* List  $F = \{o_1, \dots, o_f\}$  and pick  $q_{o_i} \in H_{o_i} \setminus \{q_{o_1}, \dots, q_{o_{i-1}}\}$ , possible since  $|H_{o_i}| \geq f > i - 1$ . Let  $x_o \in E_o$  have protecting good  $q_o$ . Define  $\sigma(o) = x_o$  for  $o \in F$ , and  $\sigma(j)$  = an out-neighbour in  $D$  for  $j \notin U \cup F$  (F2). Then  $\sigma(j) \notin U$ ,  $\sigma(j) \neq j$ , and  $j \rightarrow \sigma(j)$  is an arc of  $D^+$ . Iterating  $\sigma$  from any agent outside  $U$  repeats an agent and gives a cycle; its exposure arcs  $o \rightarrow x_o$  carry the distinct labels  $h_o$ .  $\square$

*Proof (of Theorem 4).* If (P) fails, Lemma 6(b). If  $F = \emptyset$ , Lemma 7(d) gives a cycle of  $D$ , and Lemma 6(a). Otherwise no free agent absorbs, so none holds nothing and  $|H_o| \geq |F|$  for all  $o \in F$  (Lemma 7(a,b)); Lemmas 9 and 8. Each step is a walk or a greedy choice.  $\square$

## 5 The Algorithm Draft and Exchange

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**Algorithm DE. 1. Peel.** While at least two agents remain and rule R1 applies to one of them, it leaves with its set  $P$ . If one agent remains, it takes all remaining goods; stop. **2. Draft.** The remaining agents, in index order, take the first of  $a_i, b_i, c_i$  not yet taken, or nothing. **3. Exchange.** Repeat: if (P) fails, apply the pair chain (or need cycle) of Lemma 6(b); else, if every agent holds its pair, or a free agent holds nothing, or a free  $o$  has  $|H_o| \leq |F| - 1$ , stop with that absorber (any agent if all hold pairs;

$H = \emptyset, \emptyset, H_o$  respectively); else, if  $F = \emptyset$ , apply a need cycle; else apply the exchange cycle of Lemma 9. **4. Complete** with the absorber  $o$  and  $H$ .

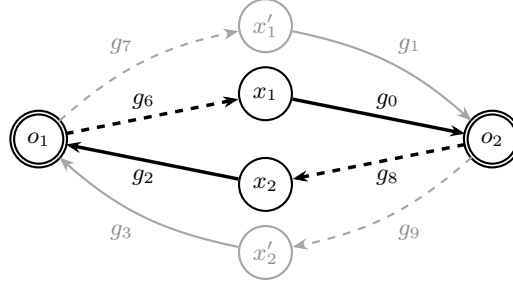
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**Theorem 5.** *On every instance with  $|R_i| \leq 3$  for all  $i$ , DE returns a complete EFX<sub>0</sub> allocation in which all bundles but at most one have at most two goods, after at most  $n$  peels and  $4n$  exchanges.*

*Proof.* By Lemma 3 and induction on the peels, it suffices that the allocation of the residual instance is EFX<sub>0</sub>; with one agent this is trivial. Otherwise, by Lemma 4, we are in the core case. The draft is valid (Lemma 5). Each exchange keeps validity and raises  $\sum_i u_i \in [0, 4n]$  by at least 1 (Lemmas 6, 7(d), 8, 9, as in the proof of Theorem 4), so there are at most  $4n$ . At the stop,  $o$  is a valid absorber with  $H$  (Lemma 7, or, if every agent holds its pair, the remark after Theorem 4), and the completion is EFX<sub>0</sub> with only  $X_o$  large (Theorem 3). Peeled agents get at most one good.  $\square$

Theorems 1 and 2 follow; each iteration needs one pass over agents and goods, so DE takes  $O(n(n+m))$  steps after reading the input (not formalized; on real values, in the comparison model of Section 1). The absorber test is applied only under (P), where it is a count; in general it would ask for a smallest hitting set.

*Example.* Goods  $g_0, \dots, g_9$ ; agents  $x_1, x'_1, x_2, x'_2, o_1, o_2$  rank  $(g_0, g_4, g_6), (g_1, g_4, g_7), (g_2, g_5, g_8), (g_3, g_5, g_9), (g_2, g_3, g_4), (g_0, g_1, g_5)$  (values 4, 3, 2, say). Nobody is peeled; the draft gives the  $x$ 's their tops,  $o_1$  its  $c = g_4$  and  $o_2$  its  $c = g_5$ . Then  $NA = \{g_0, \dots, g_3\}$ ,  $J = \{g_6, \dots, g_9\}$ ,  $F = \{o_1, o_2\}$ ,  $H_{o_1} = \{g_6, g_7\}$  (from  $x_1, x'_1$ ) and  $H_{o_2} = \{g_8, g_9\}$ : no free agent absorbs. There is no need cycle, pair chain, or cycle of  $D^+$  with one exposure arc, and each of the four valid states that dominate this one gives pairs to two agents. DE picks  $q_{o_1} = g_6$ ,  $q_{o_2} = g_8$  and exchanges along  $o_1 \rightarrow x_1 \rightarrow o_2 \rightarrow x_2 \rightarrow o_1$  (Figure 1):  $x_1$  takes  $\{g_4, g_6\}$ ,  $o_2$  takes  $g_0$ ,  $x_2$  takes  $\{g_5, g_8\}$ ,  $o_1$  takes  $g_2$ . Now  $NA = \emptyset$ ,  $J = \{g_7, g_9\}$ , and  $x'_1$  absorbs with  $H = \emptyset$ :  $X_{x'_1} = \{g_1, g_7, g_9\}$ , an EFX<sub>0</sub> allocation. One-arc exchanges are the rotations of K3ALG [10]; the single-move local search gets stuck here [11]. A state with no free absorber and no such short move needs  $n \geq 6$  and  $m \geq 8$ , both attained (a written proof, not refereed), and on a family of rings every exchange cycle needed  $k$  exposure arcs in our computations for  $k \leq 5$  (not proved; Appendix C).



**Fig. 1.** The exchange digraph of the drafted state of the example. Double circles: free agents. Dashed: exposure arcs (labelled by the protecting good); solid: need arcs (labelled by the good that moves). Black: the cycle chosen by DE.

## 6 Verification Status, Evidence and Conclusion

The Improvement Lemma, soundness and DE are written proofs, checked by computer as described here, not machine-checked. The Improvement Lemma was found twice independently: the proof above and a “rainbow walk” that leaves each free agent by an exposure arc with an unused label ([11], `k3/simplify/po/hall/NOTES.md` and `po/potential/NOTES.md`). One referee, an independent AI agent given only the written proof, found no error (`po/referee/README.md`); there has been no human peer review; its four presentation fixes are applied (explicit completion and owner constraint;  $h_x$  well defined; Lemma 6(a) when  $F = \emptyset$ ; one completion checked per state in the tests). Three implementations assert every claim of the proofs, with raw EFX<sub>0</sub> checks of the outputs, without failure: the first proof as code, on all 1,693,750 valid states of every ranking profile with  $n = 2$  ( $m \leq 6$ ),  $n = 3$  ( $m \leq 7$ ) and  $n = 4$ ,  $m = 5$ , plus samples; the second proof, on about 15.5 million valid states; and the referee’s code, written from the text alone with random choices, on about 2.9 million runs, including every profile with  $n = 2$  ( $m \leq 6$ ),  $n = 3$  ( $m \leq 7$ ) and  $n = 4$  ( $m = 5, 6$ ), and 300,000 real-valued instances with peeling. Our script `paper/k3-simple/examples/check_examples.py` recomputes every example of this paper and runs DE, peeling included, on 40,000 random instances (Appendix E). Existence, and the shape for exactly three balanced goods, are also Lean-checked, with the old proof [10]. The shape cannot be improved in the obvious ways: some core instances need a bundle of three goods, and some need four (Appendix D).

EFX<sub>0</sub> with four relevant goods per agent is open (for EFX, see [17]); so are a better bound than  $4n$  exchanges, the least size of the large bundle, and a Lean proof of the Improvement Lemma.

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## A Proofs for Section 2

*Proof (of Lemma 1).* (a)  $\theta_i(\{h\}) = 0$ . (b) The minimum is at least 0 and  $v_i(B) = v_i(B \cap R_i)$ . (c)  $v_i(g) + v_i(h) - \min(v_i(g), v_i(h))$ .  $\square$

*Proof (of Lemma 2).* Fix  $j \neq i$ ; if  $|X_j| \leq 1$ ,  $\theta_i(X_j) = 0$ . Let  $|X_j| \geq 2$  and  $T = X_j \cap R_i$ , so  $\theta_i(X_j) \leq v_i(T)$ . (a)  $T \subseteq \{a_i\}$ , and  $v_i(a_i) \leq v_i(b_i) + v_i(c_i) \leq v_i(X_i)$ . (b) If  $y = \perp$ ,  $T \subseteq G_\perp$  is empty. Otherwise each  $g \in T$  is neither in  $G_y$  nor  $y$ , so  $y \succ_i g$  and  $v_i(g) \leq v_i(y) \leq v_i(X_i)$ ; this settles  $|T| \leq 1$ . If  $|T| \geq 2$ , then  $y = a_i$  and  $T = \{b_i, c_i\}$ ; if  $|X_j| = 2$ ,  $\theta_i(X_j) = v_i(b_i) \leq v_i(a_i)$  by Lemma 1(c); if  $|X_j| \geq 3$  we are in the exception.  $\square$

*Proof (of Lemma 3).* Every other bundle lies in  $M \setminus P$ , so its threat to  $i$  is at most  $v_i(M \setminus P) \leq v_i(P)$ . The bundle  $P$  has at most one good, so its threat is 0. Other constraints are those of  $Z$ .  $\square$

*Proof (of Lemma 4).* R1 fails, so  $v_i(p) > 0$  and  $v_i(G \setminus \{p\}) > v_i(p)$ . With at most two valued goods,  $v_i(G \setminus \{p\})$  would be the value of at most one good, at most  $v_i(p)$ . So  $i$  values three goods,  $p = a_i$ , and  $v_i(b_i) + v_i(c_i) > v_i(a_i)$ .  $\square$

If  $|R_i| \leq 2$  for all agents, serial dictatorship with the last agent taking the rest is EFX<sub>0</sub>: each pick satisfies R1 (Lemma 3, induction).

## B Proof of Theorem 3

*Proof. Sizes.* A free agent other than  $o$  holds at most one good and receives at most one; a pair holder has two goods; other agents keep one good. So only  $X_o$  can have three or more goods. *(N) Needed goods stay alone.* For  $g \in NA$ , validity gives  $Y_j = \{g\}$ ;  $j \notin U \cup F$ , so  $j \neq o$  and  $j$  receives no good of  $H$ ; so  $X_j = \{g\}$ . *(OC)* No  $x \neq o$ ,  $x \notin U$ ,  $Y_x = \{a_x\}$ , has  $b_x, c_x \in X_o$ : otherwise  $b_x, c_x \in Y_o \cup J$ , so  $x \in E_o$ , and by (A2)  $H$  contains a leftover good of  $\{b_x, c_x\}$ , which went to another agent and is not in  $Y_o$ , hence not in  $X_o$ . *Safety.* Agents are balanced. If  $i \in U$ , Lemma 2(a). If  $i \notin U$ , apply Lemma 2(b) with  $y = y_i$  (or  $\perp$ ):  $G_y = N_i \subseteq NA$ , so its goods are alone by (N). In the exception,  $y_i = a_i$  and  $b_i, c_i$  lie in a bundle of at least three goods, which is  $X_o$  with  $o \neq i$ ; (OC) excludes this.  $\square$

The proof uses only validity, the definition of a valid absorber, and balance of pair holders.

## C Short Moves and Large Exchanges

A *short move* is a need cycle, a pair chain, or an exchange along a cycle of  $D^+$  with one exposure arc. The rotation of K3ALG [10] and K3S [11] is a short move: an agent  $k$  exposed for  $r$  takes its pair and its top travels along a need chain to  $r$ .

**Proposition 1** ([11], `po/potential/NOTES.md`, Lemma 3; written, not refereed). *In the core case, let  $Y$  be valid, with no free valid absorber,  $U \neq N$ , and no short move. Then  $|F| \geq 2$ ,  $n \geq 6$  and  $m \geq 8$  (and  $n \geq 9$ ,  $m \geq 12$  if  $|F| \geq 3$ ). The bounds  $n \geq 6$  and  $m \geq 8$  are attained.*

*Proof.* (P) holds (no pair chain) and  $D$  is acyclic (no need cycle), so every walk along need arcs ends at a free agent (F2). By Lemma 7,  $F \neq \emptyset$ , every free agent holds a good,  $|H_o| \geq |F|$  for  $o \in F$ , and the sets  $E_o$  are disjoint with  $|E_o| \geq |H_o|$ . Every agent holds a good. If  $F = \{o\}$ , some  $x \in E_o$  walks along need arcs to  $o$ , and  $o \rightarrow x \rightarrow \dots \rightarrow o$  is a short move. If  $F = \{o_1, o_2\}$ : walks from  $E_{o_1}$  cannot end at  $o_1$  (short move), so they end at  $o_2$ ; one of the at least two agents of  $E_{o_1}$  is not  $o_2$ , so a need arc enters  $o_2$ ,  $o_2$  needs a good, and  $o_2 \notin E_{o_1}$ . So  $E_{o_1}$ , and symmetrically  $E_{o_2}$ , hold at least two non-free agents:  $n \geq 6$ , and  $m \geq n + |J| \geq 6 + 2$ . If  $|F| = k \geq 3$ , the  $E_o$  hold at least  $k^2$  agents, at most  $k$  of them free:  $n \geq k^2$ ,  $m \geq k^2 + k$ . The example of Section 5 has  $n = 6$ ; the state with rankings  $o_1: (g_4, g_5, g_0)$ ,  $o_2: (g_6, g_7, g_1)$ ,  $x_1: (g_4, g_1, g_2)$ ,  $x'_1: (g_5, g_3, g_1)$ ,  $x_2: (g_6, g_0, g_2)$ ,  $x'_2: (g_7, g_3, g_0)$ , where  $o_1$  holds  $g_0$ ,  $o_2$  holds  $g_1$  and the  $x$ 's their tops, has  $n = 6$ ,  $m = 8$  (checked by our script).  $\square$

In the notes' family of rings with  $k$  free agents (each needing the roots of a binary need tree whose leaves are exposed for the previous free agent), every cycle of  $D^+$  with distinct labels has exactly  $k$  exposure arcs; this was checked by computer for  $k \leq 5$  (notes) and  $k \leq 4$  (our script), not proved.

## D Limits of the Shape

**Lemma 10 (cases of safety).** *Let  $R_i = \{a, b, c\}$  with  $v_i(a) > v_i(b) > v_i(c) > 0$  and  $v_i(a) < v_i(b) + v_i(c)$ . Agent  $i$  is safe in  $X$  iff (T)  $i$  holds  $a$ , and also  $b$  or  $c$ , or  $b$  and  $c$  are not together in a bundle of at least three goods; or (BC)  $i$  holds  $b$  and  $c$ ; or (B)  $i$  holds  $b$  and  $a$  is alone; or (C)  $i$  holds  $c$ , and  $a$  and  $b$  are alone; or (E)  $a, b, c$  are all alone.*

*Proof.* Write  $x$  for  $v_i(x)$ :  $0 < c < b < a < b+c < a+c < a+b < a+b+c$ . Let  $S = X_i \cap R_i$  and, for  $j \neq i$ ,  $T = X_j \cap R_i$ ;  $\theta_i(X_j)$  is 0 if  $T = \emptyset$  or  $|X_j| = 1$ ,  $v_i(T)$  if  $X_j \not\subseteq R_i$ , and the larger value if  $X_j = T$  has two goods. If  $S$  contains  $a$  and another good, every threat is at most  $b < a+c$ : safe (T). If  $S = \{b, c\}$ , every threat is at most  $a < b+c$ : safe (P). If  $S = \{a\}$ , the threat is  $b+c > a$  exactly when  $b, c$  share a bundle of at least three goods (it then has a good outside  $R_i$ ), and at most  $b$  otherwise: (T). If  $S = \{b\}$ : if  $a$  is not alone its bundle has threat at least  $a > b$ ; if  $a$  is alone the remaining threat is at most  $c$ : (B). If  $S = \{c\}$ :  $a$  must be alone, and  $b$  too (its bundle otherwise has threat at least  $b > c$ ): (C). If  $S = \emptyset$ : any bundle of two or more goods meeting  $R_i$  has positive threat: (E). Conversely each case is one of these safe situations.  $\square$

**Proposition 2 (limits of the shape).** *With values as in Lemma 10 for every agent: (a) if agents 0, 1, 2 rank  $g_0 \succ g_1 \succ p_i$ , with  $p_i$  valued only by  $i$ , no EFX<sub>0</sub> allocation has all bundles of at most two goods, and  $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$  is EFX<sub>0</sub> in any order; (b) if agents 0, 1, 2, 3 rank  $g_0 \succ g_2 \succ p_0$ ,  $g_0 \succ g_2 \succ p_1$ ,  $g_1 \succ g_2 \succ p_2$ ,  $g_1 \succ g_2 \succ p_3$ , every EFX<sub>0</sub> allocation has sizes 4, 1, 1, 1, and one exists.*

*Proof.* (a) Bundles of at most two goods have sizes 2, 2, 1: one good is alone. Two agents do not hold  $g_0$ ; (T) fails for them, and (C), (E) need two alone goods; so both are in (BC) or (B) and hold  $g_1$ : impossible. The stated allocation is (T), (B), (C). (b) Each top  $g_0, g_1$  has a different *loser*, an agent ranking it first without holding it. A loser is safe only by (BC) or (B), holding  $g_2$ , or by (C) or (E), with  $g_2$  alone. Both cannot hold  $g_2$ , so  $g_2$  is alone and (BC) is impossible; then each loser needs its top alone too. So  $g_0, g_1, g_2$  are alone in three bundles, and the fourth is  $\{p_0, \dots, p_3\}$ . Giving  $g_0, g_2, g_1$  and the  $p_i$  to agents 0, 1, 2, 3 is EFX<sub>0</sub> (cases T, B, T, C).  $\square$

Our script confirms both parts by listing all allocations.

## E Evidence in Detail

The tests ([11]; k3/simplify/po/hall/, po/potential/, po/referee/) are evidence, not proof; values 4, 3, 2 suffice for distinct strictly balanced values by Lemma 10, and the referee also used real values. *First proof as code*: every valid state of every ranking profile with  $n = 2$  ( $m \leq 6$ ),  $n = 3$  ( $m \leq 7$ ),  $n = 4$  ( $m = 5$ ): 1,693,750 states, 269,028 not completable, each improved; 183,484 sampled states; one completion per completable

state checked  $\text{EFX}_0$ ; all 774,248 Pareto-optimal states completable; DE’s loop on 44,971 profiles. *Second proof as code*: about 15.5 million valid states (every profile with  $n = 2, m \leq 6$ ;  $n = 3, m \leq 7$ ;  $n = 4, m \leq 6$ ; random  $n \leq 8$ ; cores  $n = 5, 6$ ). *Referee*: 2,006,886 profiles (every profile with  $n = 2, m \leq 6$ ;  $n = 3, m \leq 7$ ;  $n = 4, m = 5, 6$ ), 30,700 cores ( $n = 5$ ), 200,000 random ( $n \leq 9$ ), 273,805 states, 1,130,775 completions, 65,816 rejection-sampled states (with need cycles), 3,000 rings ( $k \leq 9, n \leq 304$ ), 300,000 real-valued instances with peeling. *This paper’s script* (independent): the example, the instances of Appendices C and D, the first rows of the first set (cross-checked with `hall.py`), and DE on 40,000 random instances ( $n \leq 9$ ; ties, top-heavy agents, goods nobody values): all  $\text{EFX}_0$  with at most one bundle of more than two goods, at most 8 exchanges. No test failed.

## F Related Work

For monotone valuations that are not additive, EFX allocations need not exist [2,14]. A 3-relevant instance with no complete  $\text{EFX}_0$  allocation would refute the additive EFX conjecture (perturb all values by a small  $\varepsilon > 0$ ; compare [5]), but perturbing destroys the relevance structure.

**Table 1.** Existence results for complete allocations under restrictions. *Version cited:* the published version where one exists, else arXiv. <sup>a</sup>Goods nobody values go to an arbitrary agent. <sup>b</sup>Includes additive. <sup>c</sup>Extended abstract with a proof sketch. <sup>d</sup>An AAMAS 2025 extended abstract has the degree and girth conditions. <sup>e</sup>Accepted at EC 2026. <sup>f</sup>Existence; the shape for exactly three balanced goods; K3ALG over  $\mathbb{N}$ .

| Result     | Restriction (agents, goods, who values what)                                                                                | Notion           | Complete         | Valuations              | Algorithm                                  | Version cited           |
|------------|-----------------------------------------------------------------------------------------------------------------------------|------------------|------------------|-------------------------|--------------------------------------------|-------------------------|
| [5]        | $n = 3$                                                                                                                     | EFX <sub>0</sub> | yes              | additive                | pseudo-polynomial                          | JACM 2024               |
| [7]        | at most three distinct valuations                                                                                           | EFX <sub>0</sub> | yes              | additive                | no bound stated                            | EC 2025                 |
| [15]       | $m \leq n + 3$                                                                                                              | EFX <sub>0</sub> | yes              | monotone                | none stated                                | MOR 2024                |
| [3]        | $n = 4, m \leq 9$                                                                                                           | EFX <sub>0</sub> | yes              | additive                | none stated                                | arXiv                   |
| [6]        | graphs: at most two valuers per good, two agents share at most one good                                                     | EFX <sub>0</sub> | yes              | monotone                | polynomial time                            | MOR 2026 (EC 2023)      |
| [1]        | multigraphs: at most two valuers per good                                                                                   | EFX <sub>0</sub> | yes              | cancelable <sup>b</sup> | polynomial time                            | arXiv <sup>e</sup>      |
| [16]       | multigraphs: bipartite, or at most $\lceil n/4 \rceil - 1$ neighbours per agent, or girth at least six (non-parallel edges) | EFX <sub>0</sub> | yes              | monotone                | not efficient (stated)                     | arXiv <sup>d</sup>      |
| [8]        | at most two valuers per good, $v_i(g) \in \{0, v_g\}$                                                                       | EFX <sub>0</sub> | yes              | restricted additive     | no time bound stated                       | WINE 2024               |
| [9]        | hypergraphs of girth at least four (a good joins its valuers)                                                               | EFX <sub>0</sub> | yes              | monotone                | polynomial time                            | AAAI 2026               |
| [17]       | at most four relevant goods per agent                                                                                       | EFX              | yes <sup>a</sup> | additive                | polynomial time, announced                 | AAMAS 2024 <sup>c</sup> |
| [13]       | at most two relevant goods per agent                                                                                        | EFX <sub>0</sub> | yes              | additive; monotone      | serial dictatorship; Lean-checked          | GitHub, v1.2.0          |
| [10]       | at most three relevant goods per agent                                                                                      | EFX <sub>0</sub> | yes              | additive                | polynomial time; Lean-checked <sup>f</sup> | manuscript              |
| this paper | at most three relevant goods per agent; any number of valuers per good                                                      | EFX <sub>0</sub> | yes              | additive                | polynomial time, at most $4n$ exchanges    | this paper              |